

MindTrack - Project 1

Instance Creation:

EC2

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Instance summary for i-05dbcc5daa093b83c (brain-devops-ec2)

Updated less than a minute ago

Instance ID

i-05dbcc5daa093b83c

Public IPv4 address

13.233.215.32

Instance state

Running

Private IPv4 addresses

172.31.32.58

Public DNS

ec2-13-233-215-32.ap-south-1.compute.amazonaws.com

Private IP DNS name (IPv4 only)

ip-172-31-32-58.ap-south-1.compute.internal

Instance type

t2.medium

VPC ID

vpc-0e5ed7f7a6f038b7d

Subnet ID

subnet-0d705687e998ddb97

Instance ARN

arn:aws:ec2:ap-south-1:177593955230:instance/i-05dbcc5daa093b83c

Hostname type

IP name: ip-172-31-32-58.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.233.215.32 [Public IP]

IAM role

IMDSv2

Required

Operator

Elastic IP addresses

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations.

Auto Scaling Group name

Managed

false

Security Group Creation:

EC2

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Volumes

sg-008ab89270c9f2768 - brain-devops-sg

Details

Security group name

brain-devops-sg

Security group ID

sg-008ab89270c9f2768

Description

brain-devops-security-group

VPC ID

vpc-0e5ed7f7a6f038b7d

Owner

177593955230

Inbound rules count

4 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (4)

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
	sgr-0db6cdf9d436f62b5	IPv4	HTTPS	TCP	443	0.0.0.0/0
	sgr-0de8079698f339c0f	IPv4	All TCP	TCP	0 - 65535	0.0.0.0/0
	sgr-03a93e4b705c90d89	IPv4	SSH	TCP	22	0.0.0.0/0
	sgr-0cea8aa588c8565f2	IPv4	HTTP	TCP	80	0.0.0.0/0

pem file creation and access:

The screenshot displays the AWS Management Console on the left and a terminal window on the right. The terminal shows the process of creating an SSH key pair and connecting to an Ubuntu instance.

AWS Console (Left):

- Search bar: Search
- System load: 1.61
- Usage of /: 25.9% of 6.71GB
- Memory usage: 5%
- Swap usage: 0%
- Processes: 125
- Users logged in: 1
- IPv4 address: 172.31.32.58
- Expanded Security Maintenance for Applications: 0 updates can be applied immediately.
- Enable ESM Apps to receive additional future security updates. See <https://ubuntu.com/esm> or run: `sudo pro status`
- The list of available updates is more than a week old. To check for new updates run: `sudo apt update`
- The programs included with the Ubuntu system are the exact distribution terms for each program and individual files in `/usr/share/doc/*` copyright.
- Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent applicable law.
- To run a command as administrator (user "root"), use "sudo <command>". See "man sudo_root" for details.
- ubuntu@ip-172-31-32-58:~\$

Terminal (Right):

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd ..
PS C:\WINDOWS> cd ..
PS C:\> cd Users
PS C:\Users> cd suban
PS C:\Users\suban> cd .ssh
PS C:\Users\suban> ssh -i brain-devops.pem ubuntu@13.233.215.32
The authenticity of host '13.233.215.32 (13.233.215.32)' can't be established.
ED25519 key fingerprint is SHA256:JoCHTmfGSjGM3PDVLAJKNmZfCOJycuyXFw57PYrFATO.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.233.215.32' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Feb 10 15:05:43 UTC 2026

System load: 0.23          Processes:           125
Usage of /: 26.4% of 6.71GB Users logged in:       1
Memory usage: 6%          IPv4 address for enx0: 172.31.32.58
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Feb 10 15:03:23 2026 from 13.233.177.5
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-32-58:~$
```

Install system updates:

The screenshot shows a terminal window with the following output:

```
update-initramfs: Generating /boot/initrd.img-6.14.0-1018-aws
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart acpid.service chrony.service cron.service irqbalance.service packagekit.service polkit.service rsyslog.service ssh.service udisks2.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #1: apt[1888], sshd[1029,1437]
ubuntu @ session #4: sshd[1498,1556]
ubuntu @ user manager service: systemd[1332]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-58:~$
```

Install Git:

```
Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #1: apt[1888], sshd[1029,1437]
ubuntu @ session #4: sshd[1498,1556]
ubuntu @ user manager service: systemd[1332]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-58:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-32-58:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-32-58:~$
```

Install node.js

```
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #1: sshd[1029,1437]
ubuntu @ session #4: sshd[1498,1556]
ubuntu @ user manager service: systemd[1332]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-58:~$ node -v
v18.20.8
ubuntu@ip-172-31-32-58:~$ npm -v
10.8.2
ubuntu@ip-172-31-32-58:~$
```

Install docker:

```
aws [Search] [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@tty0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #1: sshd[1029,1437]
ubuntu @ session #4: sshd[1498,1556]
ubuntu @ user manager service: systemd[1332]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-58:~$ sudo systemctl start docker
ubuntu@ip-172-31-32-58:~$ sudo systemctl enable docker
ubuntu@ip-172-31-32-58:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-32-58:~$ newgrp docker
ubuntu@ip-172-31-32-58:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
ubuntu@ip-172-31-32-58:~$
```

Install AWS CLI

```
aws [Search] [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@tty0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #1: sshd[1029,1437]
ubuntu @ session #4: sshd[1498,1556]
ubuntu @ user manager service: systemd[1332]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-58:~$ sudo systemctl start docker
ubuntu@ip-172-31-32-58:~$ sudo systemctl enable docker
ubuntu@ip-172-31-32-58:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-32-58:~$ newgrp docker
ubuntu@ip-172-31-32-58:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
ubuntu@ip-172-31-32-58:~$ curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o aws.zip
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 63.6M  100 63.6M    0     0  122M      0  --:--:-- --:--:-- --:--:-- 122M
ubuntu@ip-172-31-32-58:~$
```

```
aws [Search] [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
inflating: aws/dist/awscli/customizations/sso/index.html
creating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
ubuntu@ip-172-31-32-58:~$ unzip aws.zip
Archive:  aws.zip
replace aws/install? [y]es, [n]o, [A]ll, [N]one, [r]ename: N
ubuntu@ip-172-31-32-58:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-32-58:~$ aws --version
aws-cli/2.33.18 Python/3.13.11 Linux/6.14.0-1018-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-32-58:~$ aws configure
AWS Access Key ID [None]: AKIAS5WLMWOPN3YMG5FY
AWS Secret Access Key [None]: zMjkeQgWU5NeaCF8hCctM03ucWtELSrOE4GldoY
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-172-31-32-58:~$
```

Install kubectl:

```
aws
[Search] [Alt+S] Ask Amazon Q
Lakshmi Srinath (1775-9395-5230)
Lakshmi Srinath

inflating: aws/dist/prompt toolkit-3.0.51.dist-info/WHHEEL
inflating: aws/dist/prompt toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
ubuntu@ip-172-31-32-58:~$ unzip aws.zip
Archive:  aws.zip
replace aws/install? [y]es, [n]o, [A]ll, [N]one, [r]ename: N
ubuntu@ip-172-31-32-58:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-32-58:~$ aws --version
aws-cli/2.33.18 Python/3.13.11 Linux/6.14.0-1018-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-32-58:~$ aws configure
AWS Access Key ID [None]: AKIASSWLMWOPN3YMGsfY
AWS Secret Access Key [None]: zMjke/QgWU5NeaCF8hCctM03ucWtELSrOE4GldoY
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-172-31-32-58:~$ curl -O https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.1/2023-04-19/bin/linux/amd64/kubectl
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
51 46.9M 51 24.2M 0 0 131k 0 0:06:06 0:03:09 0:02:57 133k
60 46.9M 60 28.3M 0 0 130k 0 0:06:07 0:03:41 0:02:26 130k^C
ubuntu@ip-172-31-32-58:~$ curl -O https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.1/2023-04-19/bin/linux/amd64/kubectl
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 46.9M 100 46.9M 0 0 5675k 0 0:00:08 0:00:08 --:--:-- 7653k
ubuntu@ip-172-31-32-58:~$ chmod +x kubectl
ubuntu@ip-172-31-32-58:~$ sudo mv kubectl /usr/local/bin/
ubuntu@ip-172-31-32-58:~$
```

Install eksctl

```
aws
[Search] [Alt+S] Ask Amazon Q
Lakshmi Srinath (1775-9395-5230)
Lakshmi Srinath

You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-32-58:~$ aws --version
aws-cli/2.33.18 Python/3.13.11 Linux/6.14.0-1018-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-32-58:~$ aws configure
AWS Access Key ID [None]: AKIASSWLMWOPN3YMGsfY
AWS Secret Access Key [None]: zMjke/QgWU5NeaCF8hCctM03ucWtELSrOE4GldoY
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-172-31-32-58:~$ curl -O https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.1/2023-04-19/bin/linux/amd64/kubectl
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
51 46.9M 51 24.2M 0 0 131k 0 0:06:06 0:03:09 0:02:57 133k
60 46.9M 60 28.3M 0 0 130k 0 0:06:07 0:03:41 0:02:26 130k^C
ubuntu@ip-172-31-32-58:~$ curl -O https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.1/2023-04-19/bin/linux/amd64/kubectl
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 46.9M 100 46.9M 0 0 5675k 0 0:00:08 0:00:08 --:--:-- 7653k
ubuntu@ip-172-31-32-58:~$ chmod +x kubectl
ubuntu@ip-172-31-32-58:~$ sudo mv kubectl /usr/local/bin/
ubuntu@ip-172-31-32-58:~$ curl --silent --location \
"https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_${uname -s}_amd64.tar.gz" \
| tar xz -C /tmp
ubuntu@ip-172-31-32-58:~$ curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_${uname -s}_amd64.tar.gz" -o eksctl.tar.gz
ubuntu@ip-172-31-32-58:~$ tar -xzf eksctl.tar.gz
ubuntu@ip-172-31-32-58:~$ sudo mv eksctl /usr/local/bin/
ubuntu@ip-172-31-32-58:~$ eksctl version
0.222.0
ubuntu@ip-172-31-32-58:~$
```

Clone Git:

```
aws | Search [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
ubuntu@ip-172-31-32-58:~$ git clone https://github.com/Vennilavan12/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 8, done.
remote: Total 8 (delta 0), reused 0 (delta 0), pack-reused 8 (from 1)
Receiving objects: 100% (8/8), 100.04 KiB | 11.12 MiB/s, done.
ubuntu@ip-172-31-32-58:~$ cd Brain-Tasks-App
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$
```

Install nginx and config:

```
aws | Search [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #1: sshd[1029,1437]
ubuntu @ session #4: sshd[1498,1556]
ubuntu @ user manager service: systemd[1332]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ sudo rm -rf /var/www/html/*
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ sudo cp -r dist/* /var/www/html/
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ sudo systemctl restart nginx
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$
```

Docker build image:

```
aws | Search [Alt+S] | Asia Pacific (Mumbai) | Lakshmi Srinath (1775-9395-5230) | Lakshmi Srinath

ubuntu@ip-172-31-32-58:~$ cd Brain-Tasks-App/
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano Dockerfile
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker build -t brain-tasks-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 487.4kB
Step 1/5 : FROM nginx:alpine
alpine: Pulling from library/nginx
589002ba0eae: Pulling fs layer
bca5d04786e1: Pulling fs layer
3e2c181db1b0: Pulling fs layer
6b7b6c7061b7: Pulling fs layer
399d0898a94e: Pulling fs layer
955a8478f9ac: Pulling fs layer
6d397a54a185: Pulling fs layer
5e7756927bef: Pulling fs layer
399d0898a94e: Waiting
955a8478f9ac: Waiting
6d397a54a185: Waiting
5e7756927bef: Waiting
6b7b6c7061b7: Waiting
3e2c181db1b0: Verifying Checksum
3e2c181db1b0: Download complete
bca5d04786e1: Verifying Checksum
bca5d04786e1: Download complete
589002ba0eae: Verifying Checksum
```

Docker build success and run the container

```
aws | Search [Alt+S] | Asia Pacific (Mumbai) | Lakshmi Srinath (1775-9395-5230) | Lakshmi Srinath

--> Running in 9cd99ec2490f
--> Removed intermediate container 9cd99ec2490f
--> ddf94822c91e
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 4a02579de37d
--> Removed intermediate container 4a02579de37d
--> b8c2a6f8cc73
Successfully built b8c2a6f8cc73
Successfully tagged brain-tasks-app:latest
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker run -d -p 80:80 --name brain-app brain-tasks-app
3c0424480b2f76ab2f4351e2f940e479cd67d1c2009c61a02efcc2d62936ec73
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint brain-app (da4dba36b7cc730dd93a35fcc3ef1721f7381bba3ff22d34dd3d936aab20c173): failed to bind host port for 0.0.0.0:80:172.17.0.2:80/tcp: address already in use

Run 'docker run --help' for more information
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker run -d -p 6000:80 --name brain-app brain-tasks-app
docker: Error response from daemon: Conflict. The container name "/brain-app" is already in use by container "3c0424480b2f76ab2f4351e2f940e479cd67d1c2009c61a02efcc2d62936ec73". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker rm -f brain-app
brain-app
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker run -d -p 6000:80 --name brain-app brain-tasks-app
9fea092af518ac1625129473313e7b059328d73665bf8ad22c9659e35dd2dd91
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
9fea092af518   brain-tasks-app "/docker-entrypoint..." 20 seconds ago Up 19 seconds 0.0.0.0:6000->80/tcp, [::]:6000->80/tcp   brain-app
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$
```

ECR creation and image push:

```
aws
[Alt+S] Ask Amazon Q
Step 3/5 : COPY dist/ /usr/share/nginx/html/
--> Using cache
--> ab3435a65e42
Step 4/5 : EXPOSE 80
--> Using cache
--> ddf94822c91e
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> b8c2a6f8cc73
Successfully built b8c2a6f8cc73
Successfully tagged simple-docker-service:latest
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker images
REPOSITORY                                TAG                IMAGE ID           CREATED            SIZE
177593955230.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app    latest            b8c2a6f8cc73      4 hours ago       62.4MB
brain-tasks-app                                           latest            b8c2a6f8cc73      4 hours ago       62.4MB
simple-docker-service                                     latest            b8c2a6f8cc73      4 hours ago       62.4MB
nginx                                                      alpine            b76de378d572      5 days ago        62.1MB
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ aws ecr get-login-password --region ap-south-1 \
| docker login --username AWS --password-stdin 177593955230.dkr.ecr.ap-south-1.amazonaws.com
WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker tag simple-docker-service:latest 177593955230.dkr.ecr.ap-south-1.amazonaws.com/simple-docker-service:latest
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ docker push 177593955230.dkr.ecr.ap-south-1.amazonaws.com/simple-docker-service:latest
The push refers to repository [177593955230.dkr.ecr.ap-south-1.amazonaws.com/simple-docker-service]
05c16343c1d3: Pushed
a830b3ab3f34: Pushed
a83ac26fdfff: Pushed
7b2903554e63: Pushed
6c0e59fd138a: Pushed
aa0fc249df10: Pushed
f33bceec50d8e: Pushed
660f9a93104f: Pushed
53998a5033c3: Pushed
989e799e6349: Pushed
latest: digest: sha256:f716e317ec0179995c541e1777ad68654e1123ee8d7fc3eba7f95d017382be22 size: 2406
```

brain-track-pipeline x SimpleDockerProject x Elastic Container Res x EC2 Instance Conn x Brain Task x Brain Task x SubanalakshmiS/Bra x +

ap-south-1.console.aws.amazon.com/ecr/repositories/private/177593955230/simple-docker-service-02fd369019c5/_details?region=ap-south-1

aws [Alt+S] Q Search Asia Pacific (Mumbai) Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath

Amazon ECR > Private registry > Repositories > Images

Amazon Elastic Container Service

▼ Private registry

Repositories

simple-docker-service-02fd369019c5

Features & Settings

▼ Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

Getting started

Documentation

simple-docker-service-02fd369019c5

Summary Images Lifecycle policy Permissions Repository tags

Images (5) Info

Filter active images

☐	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
☐	latest	Image	February 11, 2026, 04:01:49 (UTC+05.5)	26.08	sha256:55f4035c...	-
☐	-	Image	February 11, 2026, 03:30:23 (UTC+05.5)	26.08	sha256:8f699a01...	-
☐	-	Image	February 11, 2026, 03:29:46 (UTC+05.5)	26.08	sha256:a1f4d182...	-
☐	-	Image	February 11, 2026, 03:19:17 (UTC+05.5)	26.08	sha256:d9622da...	-
☐	-	Image	February 11, 2026, 02:54:25 (UTC+05.5)	26.08	sha256:48e0396...	-

Create Cluster:

```
aws
[Search] [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai) Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath

ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ eksctl create cluster --name brain-track --region ap-south-1
2026-02-10 19:25:41 [i] eksctl version 0.222.0
2026-02-10 19:25:41 [i] using region ap-south-1
2026-02-10 19:25:41 [i] setting availability zones to [ap-south-1c ap-south-1a ap-south-1b]
2026-02-10 19:25:41 [i] subnets for ap-south-1c - public:192.168.0.0/19 private:192.168.96.0/19
2026-02-10 19:25:41 [i] subnets for ap-south-1a - public:192.168.32.0/19 private:192.168.128.0/19
2026-02-10 19:25:41 [i] subnets for ap-south-1b - public:192.168.64.0/19 private:192.168.160.0/19
2026-02-10 19:25:41 [i] nodegroup "ng-0106392c" will use "" [AmazonLinux2023/1.34]
2026-02-10 19:25:41 [!] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-02-10 19:25:41 [i] using Kubernetes version 1.34
2026-02-10 19:25:41 [i] creating EKS cluster "brain-track" in "ap-south-1" region with managed nodes
2026-02-10 19:25:41 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-02-10 19:25:41 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=brain-track'
2026-02-10 19:25:41 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "brain-track" in "ap-south-1"
2026-02-10 19:25:41 [i] CloudWatch logging will not be enabled for cluster "brain-track" in "ap-south-1"
2026-02-10 19:25:41 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types=(SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)) --region=ap-south-1 --cluster=brain-track'
2026-02-10 19:25:41 [i] default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2026-02-10 19:25:41 [i]
2 sequential tasks: { create cluster control plane "brain-track",
  2 sequential sub-tasks: {
    2 sequential sub-tasks: {
      1 task: { create addons },
      wait for control plane to become ready,
    },
    create managed nodegroup "ng-0106392c",
  },
}
```

Deploy to EKS:

```
aws
[Search] [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai) Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath

2026-02-10 19:37:49 [✓] all EKS cluster resources for "brain-track" have been created
2026-02-10 19:37:49 [i] nodegroup "ng-0106392c" has 2 node(s)
2026-02-10 19:37:49 [i] node "ip-192-168-56-180.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [i] node "ip-192-168-65-161.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [i] waiting for at least 2 node(s) to become ready in "ng-0106392c"
2026-02-10 19:37:49 [i] nodegroup "ng-0106392c" has 2 node(s)
2026-02-10 19:37:49 [i] node "ip-192-168-56-180.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [i] node "ip-192-168-65-161.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [✓] created 1 managed nodegroup(s) in cluster "brain-track"
2026-02-10 19:37:49 [i] creating addon: metrics-server
2026-02-10 19:37:50 [i] successfully created addon: metrics-server
2026-02-10 19:37:51 [i] kubectl command should work with "/home/ubuntu/.kube/config", try 'kubectl get nodes'
2026-02-10 19:37:51 [✓] EKS cluster "brain-track" in "ap-south-1" region is ready

ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
ip-192-168-56-180.ap-south-1.compute.internal Ready    <none>   2m58s   v1.34.2-eks-ecaa3a6
ip-192-168-65-161.ap-south-1.compute.internal Ready    <none>   2m58s   v1.34.2-eks-ecaa3a6

ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano deployment.yaml
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano service.yaml
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl apply -f deployment.yaml
deployment.apps/brain-tasks-deployment created
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl apply -f service.yaml
service/brain-tasks-service created
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
brain-tasks-service LoadBalancer  10.100.87.232    a2c18e9d0c65348a8b6cdace8087ed4-1501053483.ap-south-1.elb.amazonaws.com 80:30876/TCP    24s
kubernetes ClusterIP  10.100.0.1      <none>           443/TCP          14m

ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$
```

Create Codebuild:

The screenshot shows the AWS CodeBuild console interface for creating a new build project. The top navigation bar includes the AWS logo, a search bar, and user information for Lakshmi Srinath. The breadcrumb trail indicates the path: Developer Tools > CodeBuild > Build projects > Create build project.

Project configuration

Project name
brain-track

A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.

Project type
Select what type of project you would like to create. [Info](#)

☒ **Default project**
Create a custom CodeBuild project.

☐ **Runner project**
Create a CodeBuild managed runner for workflows in GitHub Actions, GitHub Enterprise Actions, GitLab, or Buildkite.

Additional configuration
Description, public build access, build badge, concurrent build limit, tags

Source [Add source](#)

Source 1 - Primary
Source provider
GitHub

Credential
☒ Your account is successfully connected through Secrets Manager secret. [Manage account credentials](#).
☐ Use override credentials for this project only

Git init and pushed:

```
error: remote origin already exists.
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ git remote add myrepo https://github.com/SubanalakshmiJS/brain-track-app.git
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ git remote -v
myrepo  https://github.com/SubanalakshmiJS/brain-track-app.git (fetch)
myrepo  https://github.com/SubanalakshmiJS/brain-track-app.git (push)
origin  https://github.com/Vennilavan12/Brain-Tasks-App.git (fetch)
origin  https://github.com/Vennilavan12/Brain-Tasks-App.git (push)
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ git add .
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ git commit -m "Initial commit -repo Brain Track App"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ git branch -M main
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ git push -u myrepo main
Username for 'https://github.com': SubanalakshmiJS
Password for 'https://SubanalakshmiJS@github.com':
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 2 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (16/16), 101.42 KiB | 50.71 MiB/s, done.
Total 16 (delta 0), reused 8 (delta 0), pack-reused 0
To https://github.com/SubanalakshmiJS/brain-track-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'myrepo/main'.
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$
```

CodeBuild success

Developer Tools

CodeBuild

Source • CodeCommit

Artifacts • CodeArtifact

Build • CodeBuild

Getting started

Build projects

Build project

Settings

Build history

Report groups

Report history

Compute fleets [New](#)

Account metrics

Related integrations

Jenkins

GitHub Actions [New](#)

GitLab runners [New](#)

SimpleDockerProject-02fd369019c5:100a25af-24bb-4261-8cb4-a7892b0ab881

Stop build

Debug build

Retry build

Build status

Status

Succeeded

Initiator

codepipeline/brain-track-pipeline

Build ARN

arn:aws:codebuild:ap-south-1:17759395230:build/SimpleDockerProject-02fd369019c5:100a25af-24bb-4261-8cb4-a7892b0ab881

Resolved source version

fd63c3cef9b362dbdcf8b411db34b843174190b9

Start time

Feb 11, 2026 4:01 AM (UTC+5:30)

End time

Feb 11, 2026 4:01 AM (UTC+5:30)

Build number

11

Build logs

Phase details

Reports

Environment variables

Build details

Resource utilization

Showing the last 151 lines of the build log. [View entire log](#)

Tail logs

No previous logs

1 [Container] 2026/02/10 22:31:29.294734 Running on CodeBuild On-demand

2 [Container] 2026/02/10 22:31:29.294743 Waiting for agent ping

3 [Container] 2026/02/10 22:31:29.596323 Waiting for DOWNLOAD_SOURCE

4 [Container] 2026/02/10 22:31:30.921648 Phase is DOWNLOAD_SOURCE

5 [Container] 2026/02/10 22:31:30.923529 CODEBUILD_5964D86 /codebuild/entrypoint /bin/sh -c cat /dev/null

CodePipeline Success:

brain-track-pipeline

Edit

Stop execution

Create trigger

Clone pipeline

Release change

Pipeline

Executions

Triggers

Settings

Tags

Stage

Source

24a10179-30cf-4b9a-bf71-5355fc218a29

All actions succeeded.

CodeConnections

GitHub (via GitHub App)

2 minutes ago

7bf2e513 CodeConnections: Update bui

Entry

Build_and_Deploy

24a10179-30cf-4b9a-bf71-5355fc218a29

All actions succeeded.

Docker_Build_Tag_and_Push

AWS CodeBuild

1 minute ago

7bf2e513 CodeConnections: Update bui

Horizontal

Vertical

ECR image pushed:

Amazon Elastic Container Service

▼ Private registry

Repositories

simple-docker-service-02fd369019c5

Features & Settings

▼ Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

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simple-docker-service-02fd369019c5

Summary

Images

Lifecycle policy

Permissions

Repository tags

Images (5)

Filter active images

☐	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
☐	latest	Image	February 11, 2026, 04:01:49 (UTC+05.5)	26.08	sha256:55f4035c...	-
☐	-	Image	February 11, 2026, 03:30:23 (UTC+05.5)	26.08	sha256:8f699a01...	-
☐	-	Image	February 11, 2026, 03:29:46 (UTC+05.5)	26.08	sha256:a1f4d182...	-
☐	-	Image	February 11, 2026, 03:19:17 (UTC+05.5)	26.08	sha256:d9622da...	-
☐	-	Image	February 11, 2026, 02:54:25 (UTC+05.5)	26.08	sha256:48e0396...	-

Application launched:

Brain Tasks

Organize your thoughts, simplify your life

Search tasks...

All Tasks

Pending

Completed

Priority

No tasks found!

Create your first task to get started

Cloudwatch creation:

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

/aws/codebuild/SimpleDockerProject-02fd369019c5

Actions

View in Logs Insights

Start tailing

Search log group

Log group details

Log class

Standard

ARN

arn:aws:logs:ap-south-1:177593955230:log-group:/aws/codebuild/SimpleDockerProject-02fd369019c5:

Creation time

1 hour ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Deletion protection

Off

Data protection

-

Sensitive data count

-

Custom field indexes

Configure

Transformer

Configure

Anomaly detection

Training

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Field info

Log streams (12)

By default, we only load the most recent log streams.

Delete

Create log stream

Search all log streams

Cloudwatch logs:

brain-track-pipe x SimpleDockerPr x Elastic Container x EC2 Instance Co x CloudWatch | ap x Brain Task x Brain Task x Subanalakshmi x

ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#logsV2:logs-insights\$3FqueryDetail\$3D~(end~0~start~-3600~timeType~RELATIVE~unit~seconds~editorString~-isLiveTa...

CloudWatch

Logs Insights

Query definition

5m 30m 1h 3h 12h Custom

Compare (Off)

UTC timezone

Start tailing

Logs (1.1k)

Patterns (-)

Visualization

Logs (1.1k)

Summarize results

Investigate

Export results

Add to dashboard

Showing 1054 of 1,054 records matched

1,054 records (98.9 kB) scanned in 2.3s @ 461 records/s (43.3 kB/s)

Hide histogram

200 150 100 50 0

21:50 21:55 22:00 22:05 22:10 22:15 22:20 22:25 22:30 22:35 22:40 22:45

@timestamp @message

1 2026-02-10T22:42:28.167Z #6 DONE 0.8s

2 2026-02-10T22:42:28.167Z

3 2026-02-10T22:42:28.167Z #7 [3/3] COPY dist/ /usr/share/nginx/html/

4 2026-02-10T22:42:28.167Z #7 DONE 0.1s

5 2026-02-10T22:42:28.167Z

6 2026-02-10T22:42:28.167Z #8 exporting to image

7 2026-02-10T22:42:28.167Z #8 exporting layers 0.8s done

8 2026-02-10T22:42:28.167Z #8 writing image sha256:9ca2db3f8f89cfa318a83970eca107ed68dfebcca731f3d7cbb8f51e4e82b234

9 2026-02-10T22:42:28.167Z #8 writing image sha256:9ca2db3f8f89cfa318a83970eca107ed68dfebcca731f3d7cbb8f51e4e82b234 done

Files creation

```
aws [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
2026-02-10 19:37:49 [i] node "ip-192-168-56-180.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [i] node "ip-192-168-65-161.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [i] waiting for at least 2 node(s) to become ready in "ng-0106392c"
2026-02-10 19:37:49 [i] nodegroup "ng-0106392c" has 2 node(s)
2026-02-10 19:37:49 [i] node "ip-192-168-56-180.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [i] node "ip-192-168-65-161.ap-south-1.compute.internal" is ready
2026-02-10 19:37:49 [✓] created 1 managed nodegroup(s) in cluster "brain-track"
2026-02-10 19:37:49 [i] creating addon: metrics-server
2026-02-10 19:37:50 [i] successfully created addon: metrics-server
2026-02-10 19:37:51 [i] kubectl command should work with "/home/ubuntu/.kube/config", try 'kubectl get nodes'
2026-02-10 19:37:51 [✓] EKS cluster "brain-track" in "ap-south-1" region is ready
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-56-180.ap-south-1.compute.internal Ready    <none>   2m58s  v1.34.2-eks-ecaa3a6
ip-192-168-65-161.ap-south-1.compute.internal Ready    <none>   2m58s  v1.34.2-eks-ecaa3a6
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano deployment.yaml
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano service.yaml
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl apply -f deployment.yaml
deployment.apps/brain-tasks-deployment created
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl apply -f service.yaml
service/brain-tasks-service created
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ kubectl get svc
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
brain-tasks-service                LoadBalancer 10.100.87.232  a2c18e9d0c65348a8b6bcdace8087ed4-1501053483.ap-south-1.elb.amazonaws.com 80:30976/TCP 24s
kubernetes                         ClusterIP     10.100.0.1    <none>         443/TCP    14m
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano buildspec.yml
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$ nano appspec.yml
ubuntu@ip-172-31-32-58:~/Brain-Tasks-App$
```